

ViroForge User Manual

Generating synthetic virome datasets with known ground truth

Version 0.20.0 | Handley Lab, Washington University in St. Louis

ViroForge generates synthetic virome FASTQ datasets with complete ground truth for benchmarking virome analysis pipelines. This manual covers installation, dataset generation, and interpretation of results.

Pipeline Overview

ViroForge simulates the full virome sequencing workflow, from sample preparation through read generation. Each step models a real laboratory or computational process:

- 1 Collection Selection** — Choose from 20 built-in virome collections (gut, skin, marine, etc.) or register your own custom genomes.
- 2 Contamination Injection** — Add realistic non-viral contamination matching the body site: host DNA, ribosomal RNA, bacterial background, fungal background, reagent bacteria, and PhiX spike-in.
- 3 VLP Enrichment** — Simulate virus-like particle enrichment to concentrate viruses and remove bacteria/host cells. Choose from filtration (0.1–0.45 μm pore), ultracentrifugation, or column-based kits. Skip with `--no-vlp` for bulk metagenome simulation.
- 4 Amplification Bias** — Model amplification effects (linker-based PCR, MDA with phi29, or random-primed RdAB). Each method introduces characteristic GC and length biases.
- 5 Read Simulation** — Generate paired-end Illumina reads (via InSilicoSeq) or long reads (PacBio HiFi / Nanopore via PBSIM3).
- 6 Artifact Injection** — Add realistic sequencing artifacts: adapter read-through, PCR duplicates, low-complexity sequences, and MDA chimeras.
- 7 Output** — FASTQ files with per-read ground truth labels (`source=viral` , `source=host_dna` , etc.), metadata JSON, and composition tables.
- 8 Diagnostic Report** — An HTML report with plots showing dataset composition, community structure, and quality metrics.
- 9 Benchmarking** — Score your analysis pipeline (QC, assembly, or taxonomy) against the known ground truth.

Collections

ViroForge includes 20 curated virome collections. Each has body-site-specific contamination profiles based on published literature.

Contamination profiles. The percentages in collection metadata represent the *pre-VLP* composition. After VLP enrichment, most contamination is removed (typically >90%), leaving a dataset dominated by viral reads.

Collection Details

ID	Collection	Genomes	Host	Body Site
1	Gut Virome — Adult Healthy (Western Diet)	134	Human	Gut
2	Oral Virome — Saliva (Healthy)	47	Human	Oral
3	Skin Virome — Sebaceous Sites (Healthy)	15	Human	Skin
4	Respiratory Virome — Nasopharynx (Healthy)	43	Human	Respiratory
5	Marine Virome — Coastal Surface Water	445	None	Marine
6	Soil Virome — Agricultural	290	None	Soil
7	Freshwater Virome — Lake Surface Water	200	None	Freshwater
8	Mouse Gut Virome — Laboratory (C57BL/6)	22	Mouse	Gut
9	Wastewater Virome — Urban Treatment Plant	352	Human	Wastewater
10	IBD Gut Virome (Inflammatory Bowel Disease)	90	Human	Gut
11	HIV+ Gut Virome	55	Human	Gut
12	Cystic Fibrosis (CF) Respiratory Virome	79	Human	Respiratory
13	Human Respiratory RNA Virome	53	Human	Respiratory
14	Arbovirus Environmental (Mosquito Virome)	39	None	Environmental
15	Fecal RNA Virome	46	Human	Gut
16	Vaginal Virome (Healthy)	26	Human	Vaginal
17	Blood/Plasma Virome (Healthy)	21	Human	Blood
18	Ocular Surface Virome (Healthy)	17	Human	Ocular
19	Lower Respiratory (Lung) Virome (Healthy)	31	Human	Lung
20	Urinary Virome (Healthy)	20	Human	Urinary

Contamination Composition

Each collection has a body-site-specific contamination profile. The heatmap below shows the pre-VLP composition at the *realistic* contamination level. Use the buttons to see how prep-quality contaminants (host, rRNA, reagent, PhiX) scale at different levels, while microbial background (bacterial, fungal, archaeal) stays fixed.

Pre-VLP composition at realistic (1x) contamination level. All values are percentages of total reads.

ID	Collection	Viral	Bact.	Host	rRNA	Fungal	Arch.	Reagent	PhiX
1	Gut (Western Diet)	21.9	70.0	5.0	3.0	0.5	2.0	0.5	0.1
2	Oral (Saliva)	19.9	65.0	10.0	4.0	1.0	0.5	0.5	0.1
3	Skin (Sebaceous)	52.9	25.0	15.0	3.0	3.0	0.0	1.0	0.1
4	Respiratory (Nasopharynx)	49.9	25.0	20.0	4.0	0.5	0.0	0.5	0.1
5	Marine (Coastal)	16.9	75.0	0.1	5.0	0.5	2.0	0.5	0.1
6	Soil (Agricultural)	8.4	80.0	0.1	4.0	5.0	2.0	0.5	0.1
7	Freshwater (Lake)	23.4	70.0	0.1	5.0	0.5	0.5	0.5	0.1
8	Mouse Gut (C57BL/6)	22.4	70.0	3.0	3.0	2.0	2.0	0.5	0.1
9	Wastewater (Urban)	10.9	75.0	1.0	8.0	2.0	3.0	0.5	0.1
10	IBD Gut	18.4	70.0	5.0	3.0	2.0	1.0	0.5	0.1
11	HIV+ Gut	16.4	70.0	5.0	3.0	3.0	2.0	0.5	0.1
12	CF Respiratory	12.4	55.0	25.0	5.0	2.0	0.0	0.5	0.1
13	Respiratory RNA	74.4	0.0	15.0	10.0	0.0	0.0	0.5	0.1
14	Arbovirus (Mosquito)	91.4	0.0	0.1	8.0	0.0	0.0	0.5	0.1
15	Fecal RNA	82.4	0.0	5.0	12.0	0.0	0.0	0.5	0.1
16	Vaginal (Healthy)	13.4	60.0	20.0	3.0	3.0	0.0	0.5	0.1
17	Blood/Plasma	57.3	1.0	40.0	0.5	0.1	0.0	1.0	0.1
18	Ocular Surface	69.9	10.0	15.0	2.0	0.5	0.0	2.0	0.1
19	Lung (Lower Resp.)	48.9	20.0	25.0	4.0	1.0	0.0	1.0	0.1
20	Urinary	55.4	30.0	10.0	3.0	0.5	0.0	1.0	0.1

Database Structure

ViroForge stores all data in a SQLite database (`viroforge/data/viral_genomes.db` , ~500 MB).

Table	Records	Description
<code>genomes</code>	14,423	Viral genome sequences from RefSeq
<code>taxonomy</code>	14,423	ICTV/NCBI taxonomy (71.7% family, 88.9% class coverage)
<code>body_site_collections</code>	20	Collection metadata and contamination defaults
<code>collection_genomes</code>	~2,000	Genome-collection associations with abundances

Query the database directly with SQLite:

```
sqlite3 viroforge/data/viral_genomes.db \  
"SELECT collection_id, collection_name, n_genomes  
FROM body_site_collections  
ORDER BY collection_id;"
```

Reference Files

Contamination reads are generated from bundled reference sequences:

File	Sequences	Source
<code>host_fragments.fasta</code>	48	Human genome (T2T-CHM13v2.0), 10 kb fragments
<code>rrna_representatives.fasta</code>	23	E. coli, human, gut bacteria 16S/18S/23S/28S
<code>bacterial_fragments.fasta</code>	120	Gut/oral/skin/marine bacteria from RefSeq
<code>fungus_fragments.fasta</code>	15	Common fungi (Candida, Saccharomyces, etc.)
<code>archaeal_fragments.fasta</code>	15	Gut archaea (Methanobrevibacter, etc.)
<code>phix174.fasta</code>	1	PhiX174 genome (NC_001422.1, 5,386 bp)
<code>adapters.fasta</code>	11	Illumina TruSeq and Nextera adapter sequences

Installation

Step 1: Clone and install

```
git clone https://github.com/shandley/viroforge.git  
cd viroforge  
pip install -e .
```

Step 2: Install the read simulator

```
pip install insilicoseq
```

Step 3: Build the database

```
viroforge setup-db
```

This downloads RefSeq genomes, builds the SQLite database, and populates all 20 collections. Takes a few minutes; only needed once.

Step 4: Verify

```
viroforge browse
```

```
## You should see 20 collections listed with genome counts.
```

Optional: Benchmarking support

```
pip install -e "[benchmark]"
```

Required only for assembly benchmarking (installs `mappy` for minimap2 alignment).

Optional: Long-read support

Install PBSIM3 for PacBio HiFi or Nanopore read simulation:

```
git clone https://github.com/yukiteruono/pbsim3.git
cd pbsim3 && ./configure && make && sudo make install
```

Commands

Generate a Dataset

This is the core command. It generates FASTQ files with ground truth.

```
viroforge generate \
  --collection-id 1 \
  --output data/gut_virome \
  --platform novaseq \
  --coverage 30 \
  --vlp-protocol tangential_flow \
  --seed 42
```

```
## ViroForge v0.20.0 – Generating dataset
## Collection: Gut Virome - Adult Healthy (Western Diet) (134 genomes)
## Platform: novaseq | Coverage: 30x | VLP: tangential_flow
## Amplification: linker | Contamination: realistic
##
## [1/6] Injecting contamination... done (0.8s)
## [2/6] Applying VLP enrichment... done (0.3s)
## [3/6] Applying amplification... done (0.2s)
## [4/6] Simulating reads... done (45.1s)
## [5/6] Injecting artifacts... done (2.4s)
## [6/6] Writing output... done (1.1s)
##
## Output: data/gut_virome/
## Total reads: 1,245,382 (viral: 92.3%, host: 3.1%, bacterial: 2.8%, other: 1.8%)
```

Key options

Option	Description	Default
<code>--collection-id N</code>	Which collection to use (1–20)	Required
<code>--output DIR</code>	Output directory	Required
<code>--seed N</code>	Random seed for reproducibility	42
<code>--coverage N</code>	Coverage depth	10
<code>--platform NAME</code>	novaseq, miseq, hiseq, pacbio-hifi, nanopore	novaseq
<code>--vlp-protocol NAME</code>	VLP enrichment method	tangential_flow
<code>--no-vlp</code>	Skip VLP (bulk metagenome)	—
<code>--contamination-level</code>	clean, realistic, heavy	realistic
<code>--molecule-type</code>	dna, rna	dna
<code>--adapter-rate N</code>	Adapter contamination fraction	0.03
<code>--duplicate-rate N</code>	PCR duplicate fraction	0.10
<code>--dark-matter-fraction N</code>	Unclassified viral fraction	0.30

Generate from a Preset

Presets bundle common parameter combinations:

```
viroforge generate --preset gut-standard --output data/gut
```

Preset	Description
<code>gut-standard</code>	Human gut virome, 30x, VLP enrichment

Preset	Description
gut-bulk	Gut bulk metagenome (no VLP, ~70% bacterial)
gut-vlp	Gut virome with VLP enrichment
stool-clinical	Clinical stool (high host DNA)
marine-standard	Marine coastal virome
respiratory-rna	Respiratory RNA virome
assembly-high-coverage	100x coverage for assembly benchmarking
quick-test-short	Quick test (tiny dataset, fast)
quick-test-long	Quick long-read test

Batch Generation

Generate multiple datasets from a YAML configuration:

```
viroforge batch config.yaml --parallel 4
```

Example YAML configuration:

```
datasets:
  - name: gut_10x
    collection_id: 1
    coverage: 10
    platform: novaseq
  - name: gut_30x
    collection_id: 1
    coverage: 30
    platform: novaseq
  - name: marine_10x
    collection_id: 5
    coverage: 10
    platform: novaseq

sweep:
  coverage: [10, 30, 50]
  amplification: [linker, mda]
```

When a `sweep` block is present, ViroForge generates every combination of the listed parameters. For example, 3 coverage values × 2 amplification methods = 6 datasets, one for each combination.

Benchmarking

ViroForge includes three benchmarking modules for scoring analysis pipelines against known ground truth.

QC Benchmark

ViroForge does not perform QC itself — it **scores how well your QC tool performed**. The command takes two FASTQ files as input (nothing is written):

- `--raw-reads` — the **original raw reads** generated by ViroForge (with ground truth labels)
- `--cleaned-reads` — the **cleaned reads your QC tool already produced** (not a file ViroForge creates)

ViroForge reads both files, checks which labeled reads survived your QC tool and which were removed, and reports a score. The workflow:

```
# Step 1: Generate a dataset with ground truth
viroforge generate --preset gut-standard --output data/bench

# Step 2: Run YOUR QC tool on the raw reads
fastp -i data/bench/fastq/*_R1.fastq -o cleaned_R1.fastq

# Step 3: Ask ViroForge to score what your tool did
#           (both flags are INPUTS – ViroForge reads them and prints a report)
viroforge benchmark qc \
    --raw-reads data/bench/fastq/*_R1.fastq \
    --cleaned-reads cleaned_R1.fastq
```

```
## QC Benchmark Results
## -----
## Viral retention:          97.2% (source=viral reads kept)
## Host DNA removal:        99.1% (source=host_dna reads removed)
## rRNA removal:            98.5% (source=rrna reads removed)
## Bacterial removal:       94.3% (source=bacterial_background removed)
## False positive rate:     0.8% (viral reads incorrectly removed)
## Duplicate removal recall: 89.4% (duplicate reads removed)
```

A perfect QC tool would retain 100% of `source=viral` reads and remove 100% of everything else.

Assembly Benchmark

Like QC benchmarking, ViroForge does not assemble anything — it **scores contigs your assembler already produced**. The command takes two inputs:

- `--contigs` — the **contig FASTA your assembler already produced** (e.g., from megahit, SPAdes, or metaviralSPAdes)
- `--genomes` — ViroForge's **source genome FASTA** (the true reference genomes the reads came from)
- `--ground-truth` — ViroForge's dataset metadata JSON (optional, for expected coverage and abundance accuracy)

ViroForge aligns your contigs against the known source genomes and reports how well the assembler recovered them:


```
# Step 1: Assemble with YOUR assembler
megahit -1 cleaned_R1.fastq -2 cleaned_R2.fastq -o assembly/

# Step 2: Ask ViroForge to score the assembly
#           (both flags are INPUTS – ViroForge reads them and prints a report)
viroforge benchmark assembly \
  --contigs assembly/final.contigs.fa \
  --genomes data/bench/fasta/*.fasta \
  --ground-truth data/bench/metadata/*_metadata.json
```

```
## Assembly Benchmark Results
## -----
## Genomes recovered:      67 / 90   (contigs matching a known genome)
## Mean completeness:     82.3%     (fraction of each genome covered)
## Mean identity:         99.7%     (sequence accuracy of aligned regions)
## Chimeric contigs:      3 (1.2%)  (contigs spanning multiple genomes)
## N50:                   18,432 bp
## L50:                   24
```

Taxonomy Benchmark

Same pattern — ViroForge **scores classifications your taxonomy tool already produced**. The command takes:

- `--pipeline-output` — the **output file your classifier already produced** (e.g., Kraken2 output, Centrifuge report)
- `--ground-truth` — ViroForge's dataset metadata JSON (with the taxonomy block)
- `--taxdump-dir` — NCBI taxonomy dump (so ViroForge can compare at genus/family level, not just exact taxids)

```
# Step 1: Classify with YOUR tool
kraken2 --db /path/to/db cleaned_R1.fastq > kraken_out.txt

# Step 2: Ask ViroForge to score the classifications
viroforge benchmark taxonomy \
  --pipeline-output kraken_out.txt \
  --ground-truth data/bench/metadata/*_metadata.json \
  --taxdump-dir /path/to/taxdump
```

```
## Taxonomy Benchmark Results
## -----
## Species precision:      71.2%   (correct species among all assignments)
## Species recall:         64.8%   (known species that were found)
## Family precision:       89.4%   (correct family among all assignments)
## Family recall:          83.1%   (known families that were found)
## Known viral classified: 78.3%   (reads from known genomes assigned)
## Dark matter classified: 12.1%   (unclassified genomes misassigned)
```

ViroForge auto-detects the format for Kraken2, Centrifuge, DIAMOND, and MMseqs2. For other tools, specify `--format generic` with a tab-separated file.

Other Commands

Command	Description
<code>viroforge browse</code>	Interactive TUI collection browser
<code>viroforge report DIR</code>	View a dataset report
<code>viroforge compare DIR1 DIR2</code>	Compare two datasets
<code>viroforge presets list</code>	List all presets
<code>viroforge summary DIR</code>	Show expected ranges for a dataset
<code>viroforge setup-db</code>	Build/rebuild the database
<code>viroforge web</code>	Launch the web interface (local use only)

Understanding the Output

After generation, the output directory contains:

File	Description
fastq/*_R1.fastq	Forward reads
fastq/*_R2.fastq	Reverse reads (paired-end)
fasta/*.fasta	Source genomes and contaminants
metadata/*_metadata.json	Complete ground truth (v1.1 schema)
metadata/*_composition.tsv	Genome abundance table
metadata/*_manifest.tsv	Artifact manifests (adapters, duplicates, low-complexity)
generation_report.html	Diagnostic report with plots

Ground Truth Labels

Every read header contains a `source=` tag identifying its origin:

```
@read_00001 source=viral genome=NC_001422.1 pos=1234 strand=+
ATCGATCGATCGATCGATCGATCGATCG...
+
IIIIIIIIIIIIIIIIIIIIIIIIIIIIIIII...
```

Label	Meaning
<code>source=viral</code>	Viral genome read
<code>source=host_dna</code>	Host DNA contamination
<code>source=rrna</code>	Ribosomal RNA
<code>source=bacterial_background</code>	Bacterial contamination
<code>source=fungal_background</code>	Fungal contamination
<code>source=archaeal_background</code>	Archaeal contamination
<code>source=phix</code>	PhiX spike-in control
<code>source=reagent_bacteria</code>	Lab reagent contamination
<code>source=dark_matter</code>	Unclassified viral genome
<code>source=erv_endogenous</code>	Endogenous retroviral element (HERV)
<code>source=erv_exogenous</code>	Exogenous retrovirus (Retroviridae)

Benchmarking with labels. These source tags are the answer key for QC benchmarking. A perfect QC tool would retain all `source=viral` reads and remove everything else. The `viroforge benchmark qc` command automates this comparison.

Interpreting the Diagnostic Report

Open `generation_report.html` in your browser after generating a dataset. It contains the following sections:

Composition Pie Chart

Shows the fraction of reads from each source type (viral, host DNA, rRNA, bacterial, etc.) after VLP enrichment. A well-enriched VLP virome should be >90% viral.

Before vs. After VLP

Side-by-side bar chart showing how VLP enrichment changed the composition. Before VLP: mostly bacterial. After VLP: mostly viral. This is the primary purpose of VLP enrichment.

Top Viral Families

Horizontal bar chart showing the most abundant viral families. Check if the community structure makes sense for the body site (e.g., gut should be dominated by Caudovirales phages).

Rank-Abundance Curve

Each dot represents one viral genome. The X axis is rank (1 = most abundant), the Y axis (log scale) is the read fraction from that genome.

- **Steep curve** — A few viruses dominate, most are rare (typical of real viromes)
- **Flat curve** — All viruses equally abundant (unusual in nature)

GC Content Distribution

Histogram of GC content across all genomes. A skewed distribution could indicate amplification bias affecting the community.

Genome Length vs. Abundance

Scatter plot checking if genome size correlates with abundance. In an unbiased dataset there should be no trend. A visible pattern could indicate VLP size bias or amplification effects.

VLP Contamination Reduction Table

Shows the removal efficiency for each contamination type. Typical values: >95% host DNA removal, >90% bacterial removal, ~40% PhiX loss (small virus, passes through filters).

Artifact Summary Table

Shows how many reads were modified by each artifact injection step (adapters, duplicates, low-complexity, chimeras).

Use Case: IBD Gut Virome Analysis

This walkthrough demonstrates a complete ViroForge workflow using Collection 10 (IBD Gut Virome) — from dataset generation through QC, assembly, and taxonomy benchmarking. The IBD virome is a good test case because it has a distinctive dysbiotic composition that differs from healthy gut.

Background

Inflammatory Bowel Disease is associated with dramatic shifts in the gut virome (Norman et al. 2015, *Cell*; Clooney et al. 2019, *Cell Host & Microbe*). Key features include:

- **Reduced crAssphage-like viruses** — The dominant phage group in healthy gut is depleted ~44% in IBD
- **Expanded temperate phages** — Prophage induction during inflammation leads to blooms of Drexelviriidae, Demerecviridae, and Ackermannviridae

- **Severely skewed abundance** — A few viral species dominate, creating a steeper rank-abundance curve than healthy gut
- **Eukaryotic virus enrichment** — Mucosal barrier damage may increase Adenoviridae and Picornaviridae

Collection 10 models these features with 90 curated genomes across 5 viral groups:

Viral Group	Key Families	Genomes	IBD Relevance
crAssphage-like (reduced)	Intestiviridae, Suoliviridae, Steigviridae, Crevaviridae	20	44% reduction vs. healthy
Temperate phages (increased)	Drexelviridae, Demerecviridae, Ackermannviridae	35	Dominant group (~39%)
Microviridae	Microviridae (coliphage)	15	Altered but maintained
Inoviridae	Inoviridae (filamentous)	12	Reduced diversity
Eukaryotic viruses	Adenoviridae, Picornaviridae, Anelloviridae, Parvoviridae	8	Increased from mucosal damage

Step 1: Generate the Dataset

Generate an IBD gut virome dataset with VLP enrichment at 30x coverage:

```
viroforge generate \  
  --collection-id 10 \  
  --output data/ibd_virome \  
  --platform novaseq \  
  --coverage 30 \  
  --vlp-protocol tangential_flow \  
  --contamination-level realistic \  
  --seed 42
```

```
## ViroForge v0.20.0 – Generating dataset
## Collection: IBD Gut Virome (Inflammatory Bowel Disease) (90 genomes)
## Platform: novaseq | Coverage: 30x | VLP: tangential_flow
## Amplification: linker | Contamination: realistic
##
## [1/6] Injecting contamination... done (0.6s)
##         host_dna: 5.0% | rrna: 3.0% | bacterial: 70.0%
##         fungal: 2.0% | archaeal: 1.0% | reagent: 0.5% | phix: 0.1%
## [2/6] Applying VLP enrichment... done (0.2s)
##         Viral fraction: 85.2% → 93.3% (tangential_flow, 0.2 µm)
##         Host DNA removal: 96.5% | Bacterial removal: 91.1%
## [3/6] Applying amplification... done (0.1s)
## [4/6] Simulating reads... done (38.7s)
## [5/6] Injecting artifacts... done (1.9s)
## [6/6] Writing output... done (0.8s)
##
## Output: data/ibd_virome/
## Total reads: 1,245,382
## Composition: viral 93.3% | bacterial 6.2% | host 0.2% | other 0.3%
```

VLP pore size matters. With the default 0.2 µm tangential flow, you get ~93% viral reads. Switching to `--vlp-protocol tangential_flow_045` (0.45 µm pore) drops viral purity to ~85% because more bacteria pass through the larger pores. This models a real experimental trade-off: larger pores recover more large viruses but also more bacterial contamination.

For comparison, generate a bulk metagenome (no VLP enrichment) from the same collection:

```
viroforge generate \
  --collection-id 10 \
  --output data/ibd_bulk \
  --platform novaseq \
  --coverage 30 \
  --no-vlp \
  --seed 42
```

```
## Composition: viral 4.1% | bacterial 71.8% | host 5.0% | other 19.1%
## (No VLP enrichment – bulk metagenome composition)
```

The contrast illustrates what VLP enrichment does: in a bulk stool metagenome, bacteria dominate (~72%), and viruses are a small minority (~4%). VLP enrichment inverts this ratio.

Step 2: Inspect the Output

Check the generated files:

```
ls data/ibd_virome/
```

```
## fastq/  
## fasta/  
## metadata/  
## generation_report.html
```

View the dataset report:

```
viroforge report data/ibd_virome
```

Or open `data/ibd_virome/generation_report.html` in a browser for the full diagnostic report with plots.

Inspect ground truth labels in the FASTQ:

```
head -8 data/ibd_virome/fastq/*_R1.fastq
```

```
## @read_000001 source=viral genome=NC_048087.1 pos=4521 strand=+  
## ATCGATCGATCGATCGATCGATCGATCGATCGATCG...  
## +  
## IIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIII...  
## @read_000002 source=viral genome=NC_028857.2 pos=18203 strand=-  
## GCTAGCTAGCTAGCTAGCTAGCTAGCTAGCTAGCTAG...  
## +  
## IIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIII...
```

Check the composition table for the top viral genomes:

```
head -10 data/ibd_virome/metadata/*_composition.tsv
```

## genome_id	genome_name	family	abundance
## NC_048087.1	Klebsiella phage KOX1	Drexlerviridae	0.1250
## NC_001405.1	Human adenovirus 2	Adenoviridae	0.0620
## NC_047813.1	Faecalibacterium phage FP_oengus	Unknown	0.0530
## NC_001420.1	Genome of phage G4 (coliphage)	Microviridae	0.0500
## NC_028857.2	Faecalibacterium phage FP_Toutatis	Unknown	0.0330
## NC_019445.1	Enterobacteria phage CAjan	Drexlerviridae	0.0310
## NC_042045.1	Bacteroides phage B40-8	Suoliviridae	0.0280
## NC_048018.1	Clostridium phage phi8074-B1	Demerecviridae	0.0260
## NC_024711.1	CrAssphage	Intestiviridae	0.0240

Dysbiotic signature visible. Note how Drexlerviridae (temperate phages) dominate the top ranks, while crAssphage (Intestiviridae), which is typically the most abundant phage in healthy gut, is ranked lower. This matches the expected IBD virome dysbiosis pattern.

Step 3: Run QC and Benchmark

Run a QC tool on the raw reads, then score its performance against ViroForge's ground truth:

```
# Run fastp for quality control
fastp \
  -i data/ibd_virome/fastq/*_R1.fastq \
  -I data/ibd_virome/fastq/*_R2.fastq \
  -o cleaned_R1.fastq \
  -O cleaned_R2.fastq \
  --detect_adapter_for_pe \
  --cut_right \
  --qualified_quality_phred 20
```

```
# Score QC performance against ground truth
viroforge benchmark qc \
  --raw-reads data/ibd_virome/fastq/*_R1.fastq \
  --cleaned-reads cleaned_R1.fastq
```

```
##
## |          QC Benchmark Results          |
## |-----|
## | Viral retention:           97.2%      |
## | Host DNA removal:         99.1%      |
## | rRNA removal:             98.5%      |
## | Bacterial removal:        94.3%      |
## | Fungal removal:           91.7%      |
## | False positive rate:      0.8%       |
## | Duplicate removal recall: 89.4%      |
## |-----|
## | Read-name match rate:     99.97%     |
## | Overall F1 (viral):       0.984      |
## |-----|
##
## Report saved: qc_benchmark_report.json
```

The key metrics: **viral retention** (are you keeping the good reads?) and **contamination removal** (are you removing the bad ones?). A perfect tool scores 100% on both. The read-name match rate confirms that ViroForge's ground truth labels are correctly linking input to output reads.

Step 4: Assembly and Taxonomy Benchmarking

Assemble the cleaned reads and benchmark the results:

```
# Assemble with megahit
megahit \
  -1 cleaned_R1.fastq \
  -2 cleaned_R2.fastq \
  -o assembly/ \
  --min-contig-len 1000

# Score assembly against ground truth
viroforge benchmark assembly \
  --contigs assembly/final.contigs.fa \
  --genomes data/ibd_viroome/fasta/*.fasta \
  --ground-truth data/ibd_viroome/metadata/*_metadata.json
```

##		
##	Assembly Benchmark Results	
##		
##	Genomes recovered:	67 / 90 (74.4%)
##	Mean completeness:	82.3%
##	Mean identity:	99.7%
##	Chimeric contigs:	3 (1.2%)
##	Assembly N50:	18,432 bp
##	Assembly L50:	24
##		

Run a taxonomic classifier and benchmark it:

```
# Classify with Kraken2
kraken2 \
  --db /path/to/kraken2_viral_db \
  --paired cleaned_R1.fastq cleaned_R2.fastq \
  --output kraken_out.txt \
  --report kraken_report.txt

# Score taxonomy against ground truth
viroforge benchmark taxonomy \
  --pipeline-output kraken_out.txt \
  --ground-truth data/ibd_viroome/metadata/*_metadata.json \
  --taxdump-dir /path/to/taxdump
```

##	Taxonomy Benchmark Results	
##	Read-level:	
##	Species precision:	71.2%
##	Species recall:	64.8%
##	Species F1:	0.678
##	Family precision:	89.4%
##	Family recall:	83.1%
##	Family F1:	0.861
##	Known vs dark matter:	
##	Known viral classified:	78.3%
##	Dark matter classified:	12.1%
##		

Dark matter is the hard part. The 30% dark matter fraction (unclassified viral genomes with `family='Unknown'`) is intentionally difficult to classify. A good classifier should achieve high recall on known viruses while correctly reporting dark matter reads as unclassified rather than misassigning them. The stratified report lets you evaluate both cases separately.

Compare across tools by generating a single dataset and running multiple classifiers:

```
# Same dataset, different classifiers
viroforge benchmark taxonomy --pipeline-output kraken_out.txt --ground-truth metadata.json --taxdump-di
viroforge benchmark taxonomy --pipeline-output centrifuge_out.tsv --ground-truth metadata.json --taxdump-d
viroforge benchmark taxonomy --pipeline-output diamond_out.tsv --ground-truth metadata.json --taxdump-di

# Compare the reports
viroforge compare \
  kraken_benchmark.json \
  centrifuge_benchmark.json \
  diamond_benchmark.json
```

This is the core ViroForge workflow: generate a dataset with known ground truth, run your analysis tools, and get objective performance metrics. The IBD collection is particularly useful for benchmarking because its dysbiotic composition challenges tools differently than a healthy gut virome — the skewed abundance distribution and temperate phage blooms test how well tools handle uneven coverage.

Quick Reference

Key Defaults

Parameter	Default
Platform	NovaSeq (Illumina)
Coverage	10x
VLP protocol	Tangential flow filtration (0.2 μm)
Contamination level	Realistic (1x baseline)
Amplification	Linker-based (set via preset or generation script)
Adapter rate	3%
Duplicate rate	10%
Low-complexity rate	0.5%
Dark matter fraction	30%
Seed	42

Common Workflows

Standard gut virome dataset

```
viroforge generate --preset gut-standard --output data/gut
```

Bulk metagenome (no VLP enrichment)

```
viroforge generate --collection-id 1 --no-vlp \  
  --output data/gut_bulk --coverage 30
```

Long-read PacBio HiFi dataset

```
viroforge generate --collection-id 1 \  
  --platform pacbio-hifi \  
  --output data/gut_hifi --coverage 15
```

Hybrid assembly (matched short + long reads)

```
python scripts/generate_hybrid_dataset.py \  
  --collection-id 1 \  
  --output data/gut_hybrid \  
  --short-platform novaseq \  
  --long-platform pacbio-hifi \  
  --coverage 30 --depth 15 --seed 42
```

RNA virome dataset

```
viroforge generate --preset respiratory-rna \  
  --output data/respiratory_rna
```

Full benchmarking pipeline

```
# 1. Generate dataset with ground truth  
viroforge generate --preset gut-standard --output data/bench  
  
# 2. Run your QC tool  
fastp -i data/bench/fastq/*_R1.fastq \  
  -I data/bench/fastq/*_R2.fastq \  
  -o cleaned_R1.fastq -O cleaned_R2.fastq  
  
# 3. Score QC performance  
viroforge benchmark qc \  
  --raw-reads data/bench/fastq/*_R1.fastq \  
  --cleaned-reads cleaned_R1.fastq  
  
# 4. Assemble  
megahit -1 cleaned_R1.fastq -2 cleaned_R2.fastq -o assembly/  
  
# 5. Score assembly  
viroforge benchmark assembly \  
  --contigs assembly/final.contigs.fa \  
  --genomes data/bench/fasta/*.fasta \  
  --ground-truth data/bench/metadata/*_metadata.json  
  
# 6. Classify  
kraken2 --db /path/to/db cleaned_R1.fastq > kraken_out.txt  
  
# 7. Score taxonomy  
viroforge benchmark taxonomy \  
  --pipeline-output kraken_out.txt \  
  --ground-truth data/bench/metadata/*_metadata.json \  
  --taxdump-dir /path/to/taxdump
```

Getting Help

```
viroforge --help
viroforge generate --help
viroforge benchmark --help
```